GROUP 14

A GENERAL ANOMALY DETECTION FRAMEWORK FOR FLEET-BASED CONDITION MONITORING OF MACHINES

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PROJECT FOCUS:

Early fault detection in machine fleets before failure occurs

KEY INNOVATION:

- Fleet-based approach leveraging similar machines
- Unsupervised anomaly detection without historical labeled data
- Real-time online monitoring with visualization

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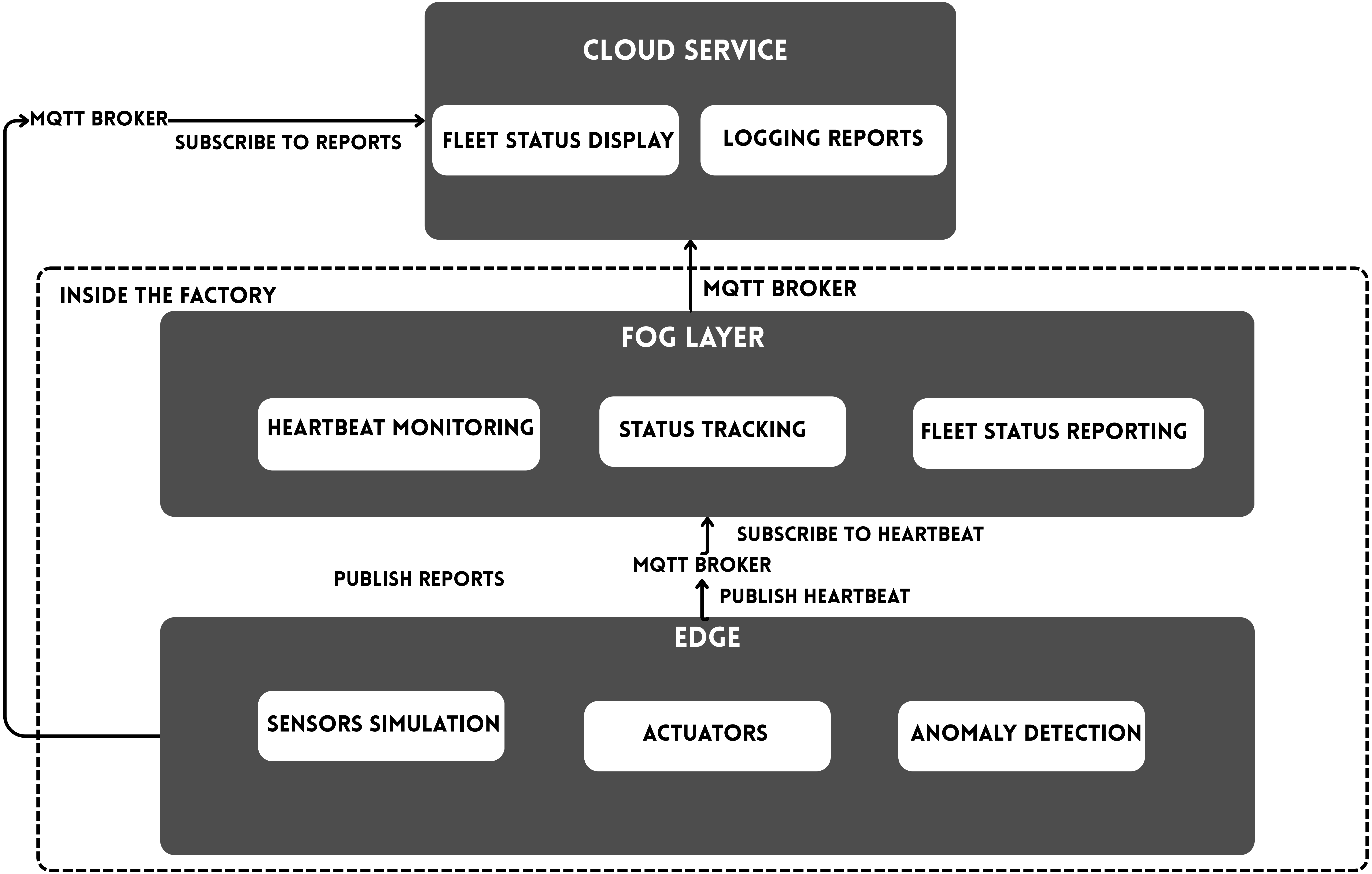
PROBLEM STATEMENT

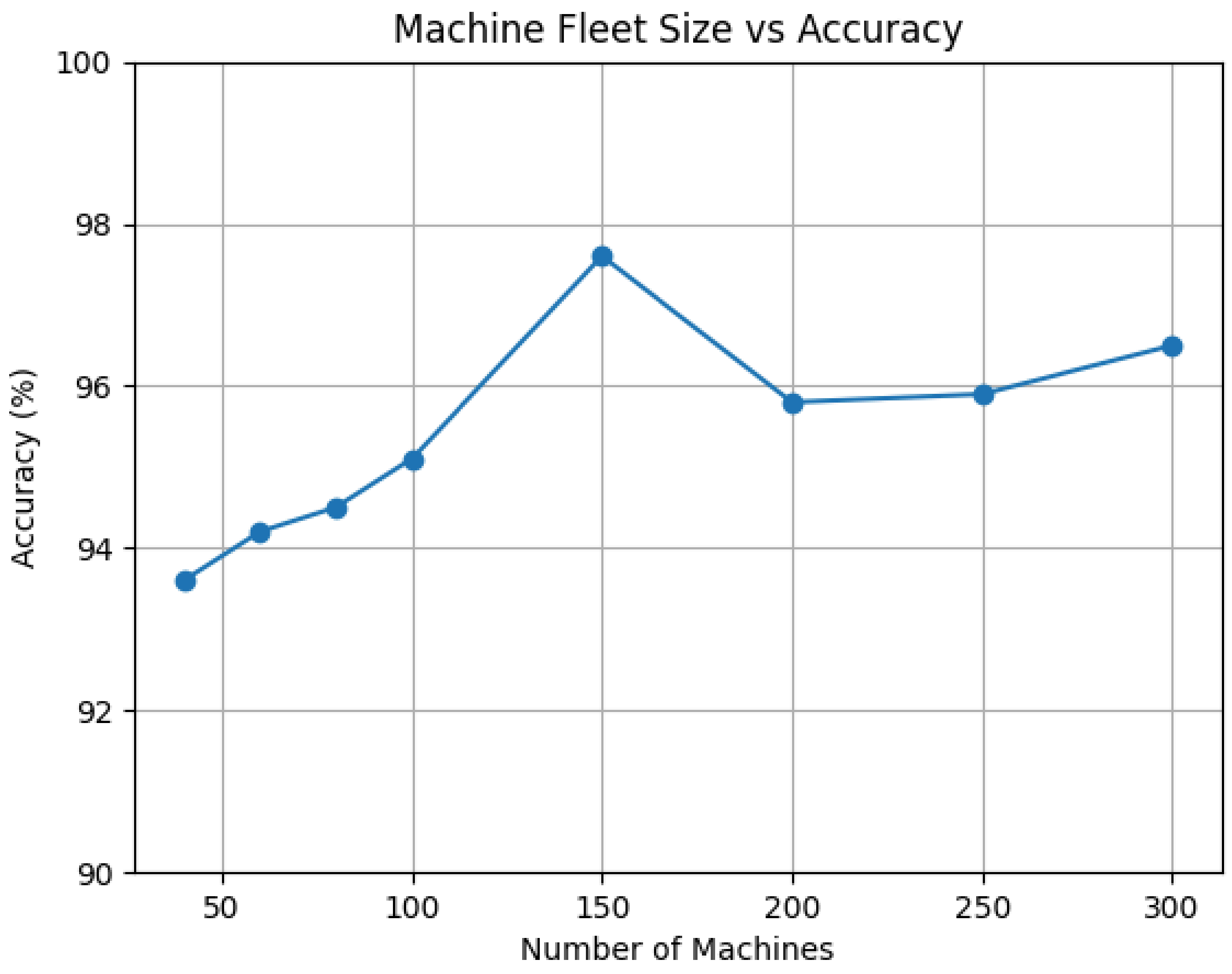
Machine failures decrease up-time and can lead to extra repair costs or even to human casualties. Recent condition monitoring techniques use artificial intelligence in an effort to avoid time-consuming manual analysis and handcrafted feature extraction. Many of these only analyze a single machine and require a large historical data set. In practice, this can be difficult and expensive to collect.

Processing all the data at the cloud will require the data to be transmitted to cloud and then process it, thereby increasing the total latency to identify the fault.

Hence, we designed a edge-fog-cloud architecture that brings the fault-detection to the edge node, and a unsupervised model that removes the need for large historical dataset.


CLOUD SERVICE
FLEET STATUS DISPLAY
LOGGING REPORTS
MQTT BROKER
SUBSCRIBE TO REPORTS
INSIDE THE FACTORY
MQTT BROKER
FOG LAYER
HEARTBEAT MONITORING
STATUS TRACKING
FLEET STATUS REPORTING
SUBSCRIBE TO HEARTBEAT
MQTT BROKER
PUBLISH REPORTS
PUBLISH HEARTBEAT
EDGE
SENSORS SIMULATION
ACTUATORS
ANOMALY DETECTION

ACCURACY

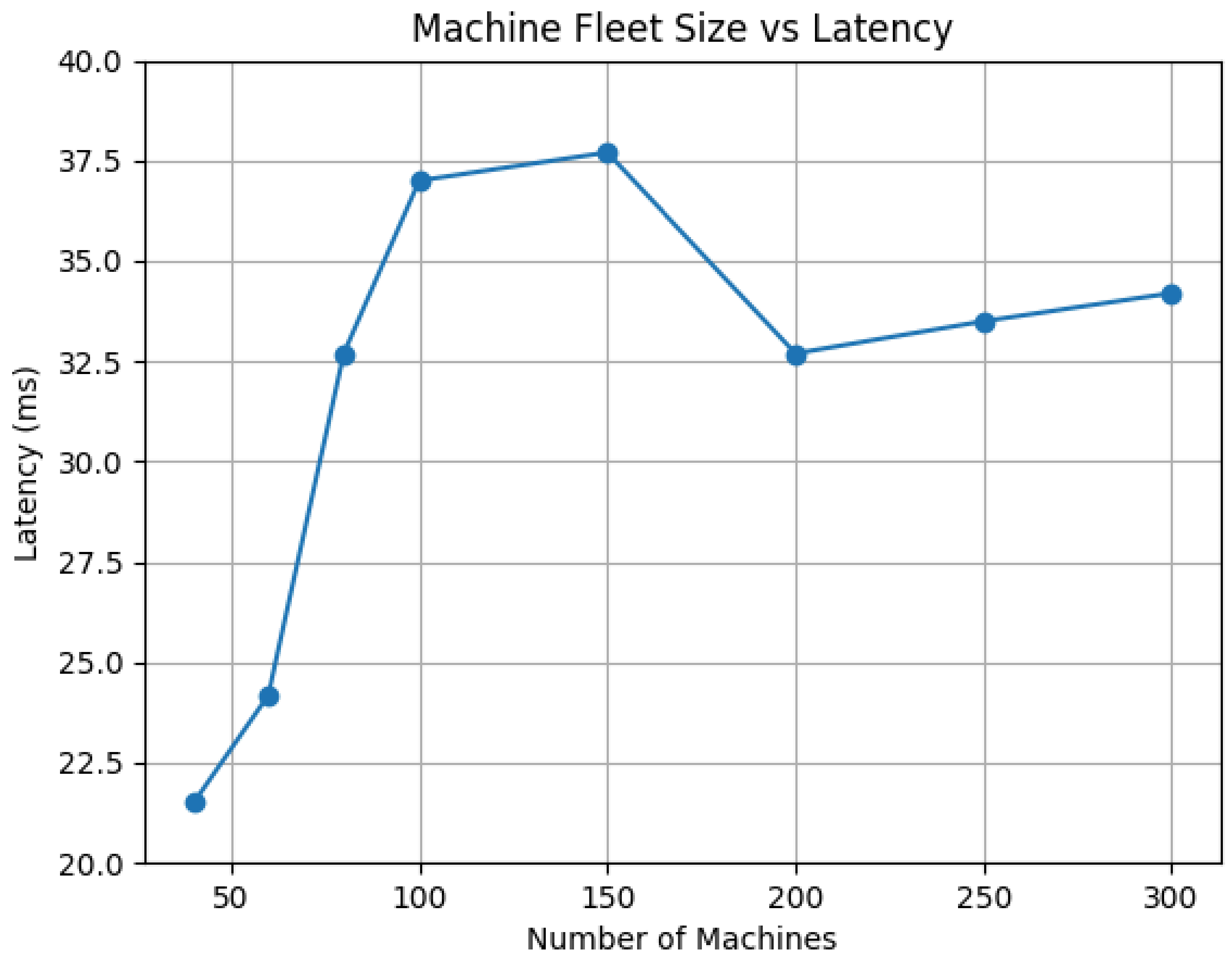

Machine Fleet Size vs Accuracy
100
98
96
94
92
90
Accuracy (%)
50
100
150
200
250
300
Number of Machines

END-END LATENCY

Size of a message for 100 machines = 180KB
Transmition rate of network = 100Mbps

Transmission time required = 14.76ms

(Propagation delay is negligible compared to transmission delay)



PROCESSING POWER

Edge:

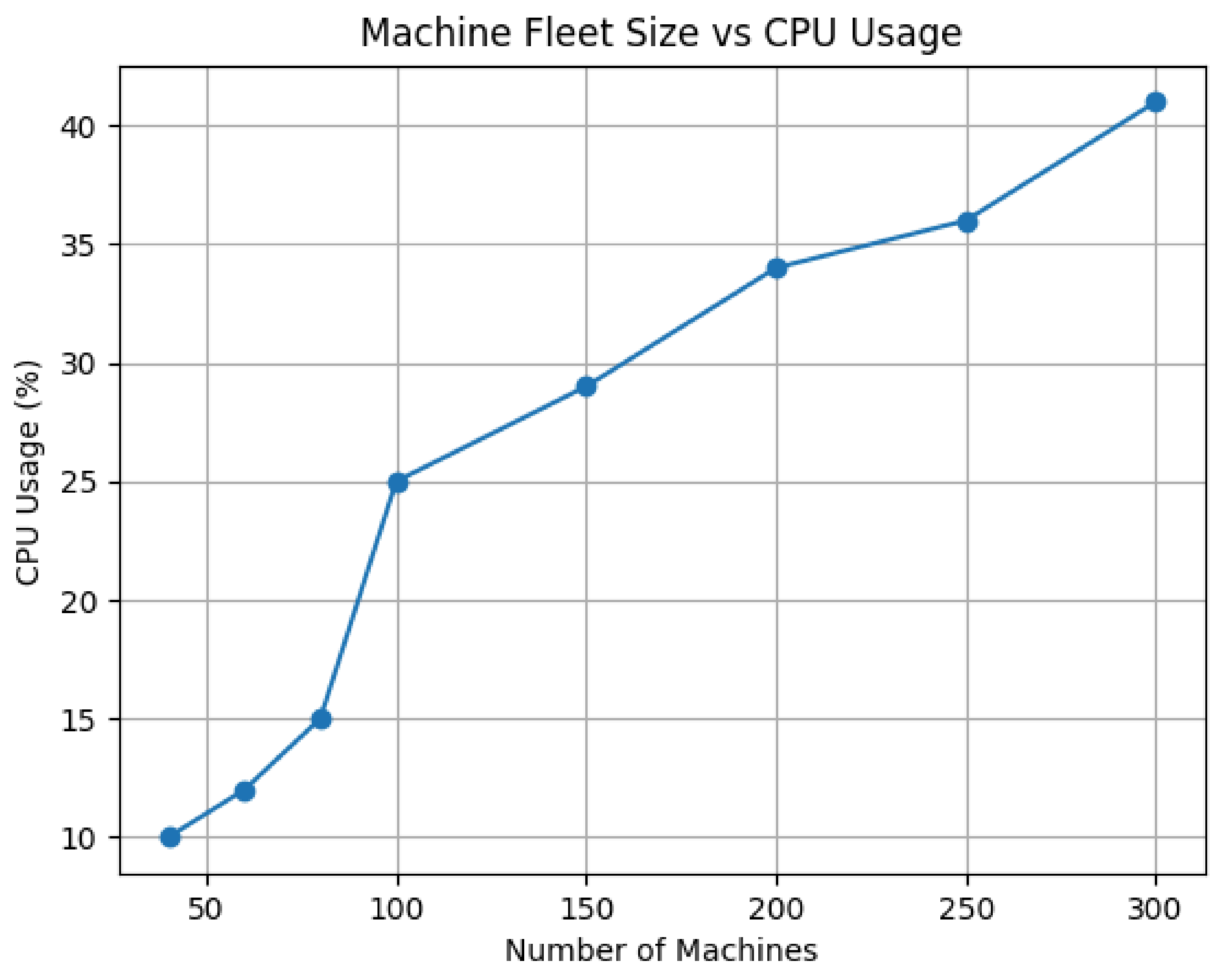
- 60-70MB is consumed by docker.
- 0.5 CPU time of the host machine.

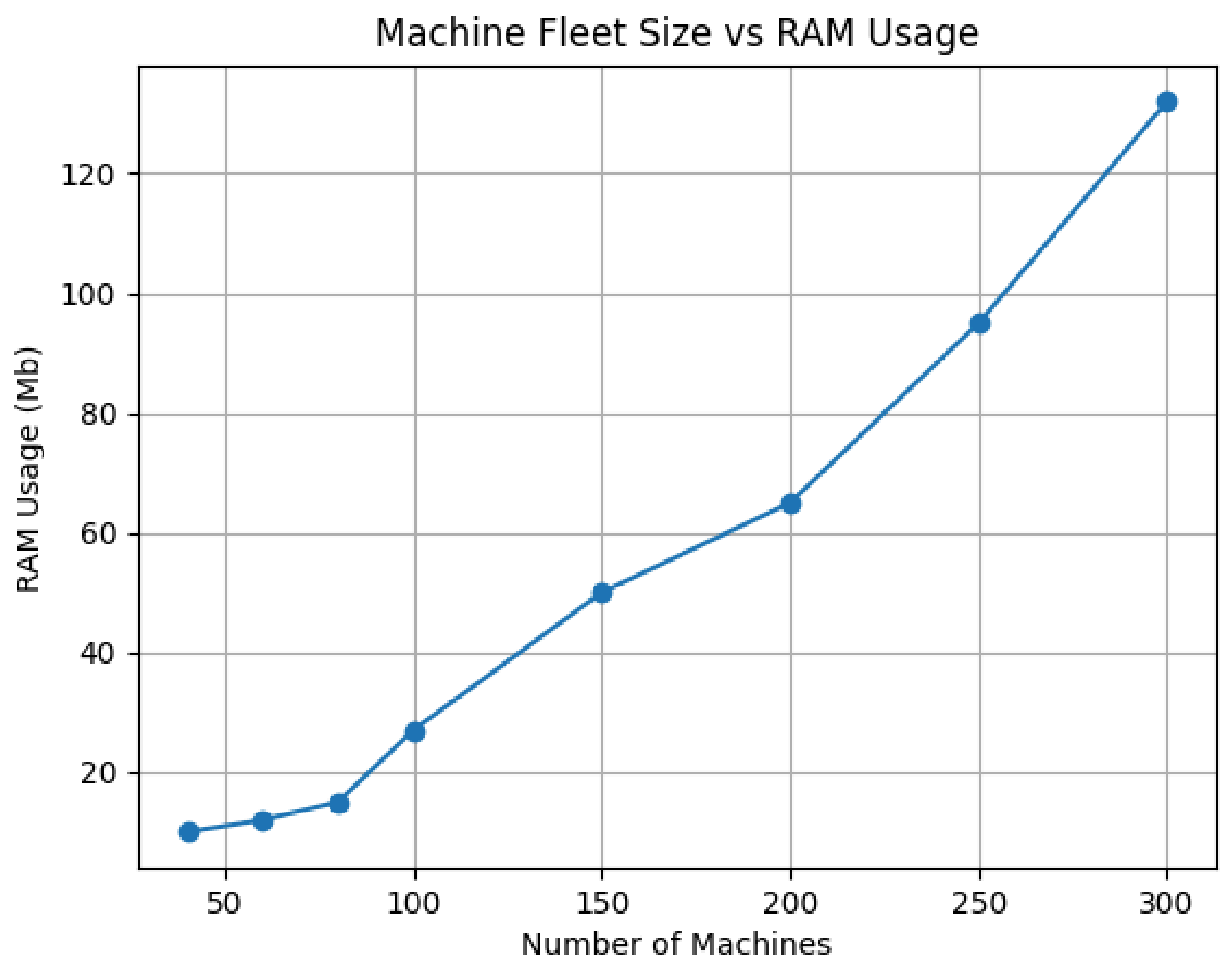
Fog:

- 512MB RAM allocated.
- 1 CPU time of the host machine.

Cloud:

- No limitations on RAM and CPU is imposed.





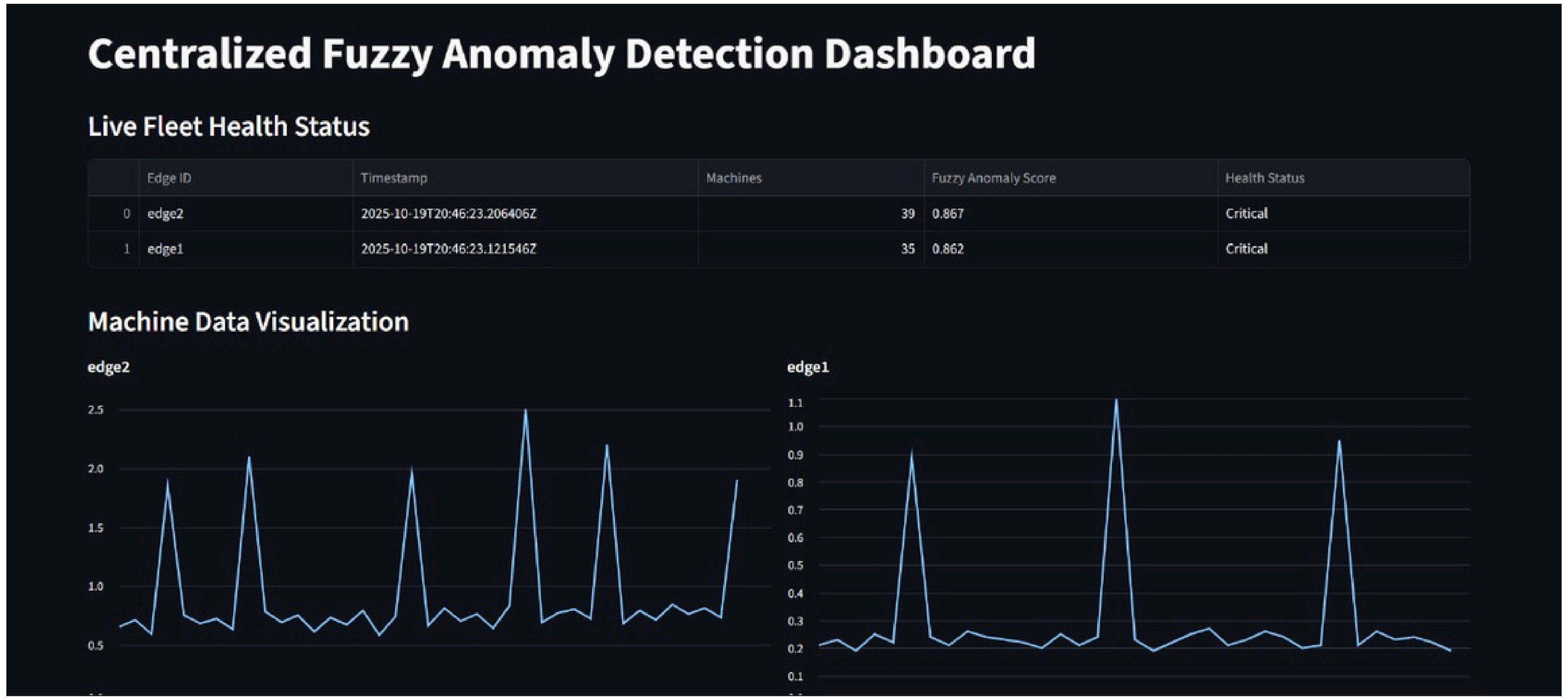
ARCHITECTURE & DATA FLOW

- **Centralized Design:** All intelligence resides in a single cloud service .
- **Edge Nodes:** Act as simple data forwarders, streaming raw sensor data via MQTT .
- **One-Way Flow:** Data moves from the edge directly to the cloud for analysis.

FUZZY INFERENCE SYSTEM (FIS) LOGIC

- **Inputs:** The system uses two key metrics calculated from the data :
  - **Max Z-Score (Severity of spikes) .**
  - **Mean Z-Score (Overall stability) .**
- **Rules:** It applies intuitive, human-readable rules like:
  - IF MaxZScore is High OR MeanZScore is Erratic THEN Anomaly is Critical .
- **Output:** A nuanced AnomalyScore between 0 (Healthy) and 1 (Critical).

FUZZY PROTOTYPE: RESULTS & LIMITATIONS



- **Centralization Bottlenecks:** Streaming all raw data consumed significant bandwidth and created a single point of failure .
- **Static Rules:** The fuzzy rules were not adaptive and required manual re-tuning for new machine behaviors or fault types .
- **Federation Incompatibility:** The design was inherently incompatible with the project's goal of creating a privacy-preserving, federated system.

ME-FLEET: ARCHITECTURE & METHODOLOGY

Edge (Local Training)
Heterogeneous devices (e.g., microcontrollers, servers) perform local training on a shared neural network.

→

Fog (Global Aggregator)
A central node performs layer-wise Federated Averaging (FedAvg) to combine model updates from all participating edge devices.

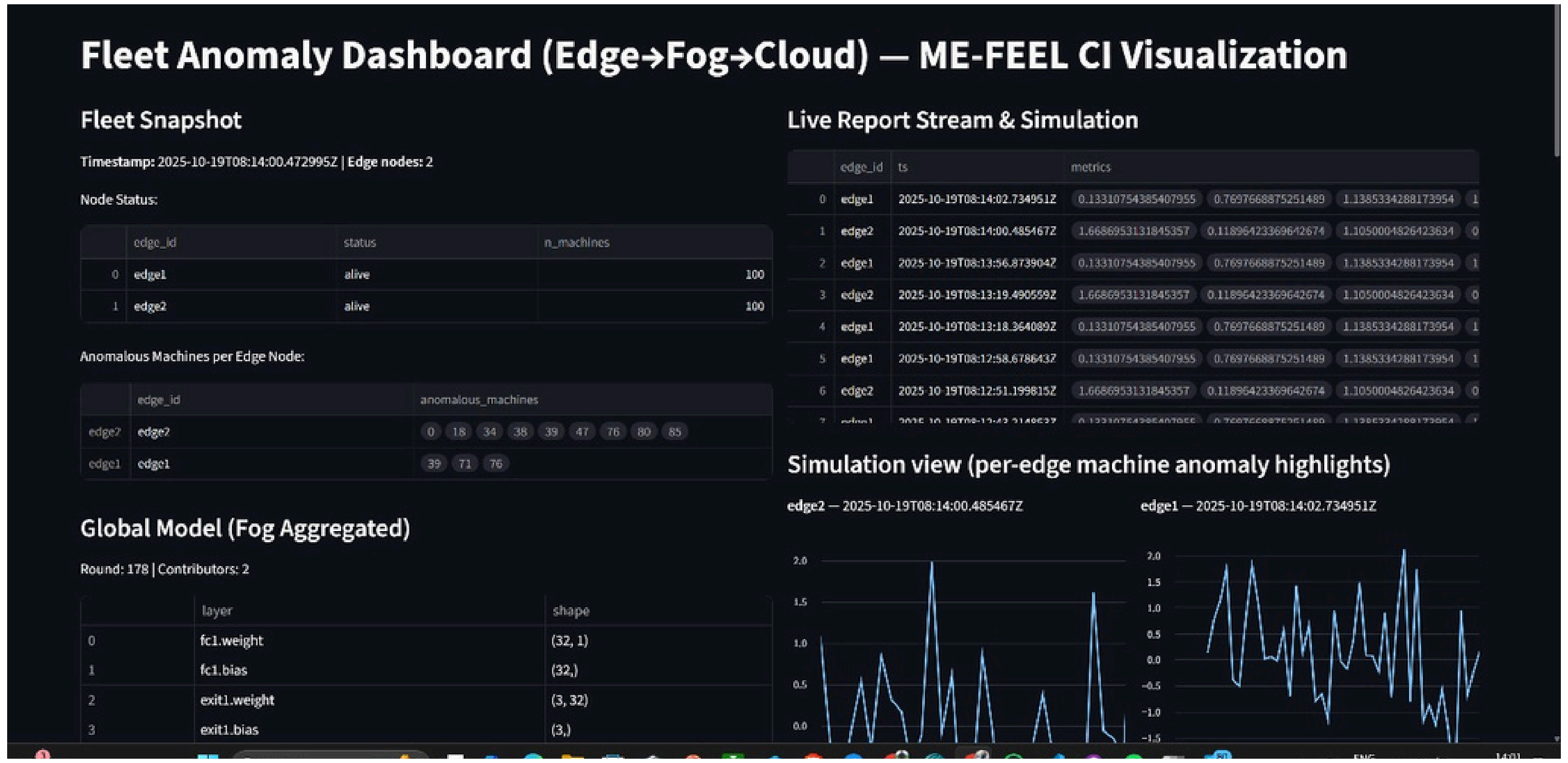
→

Cloud (Visualization)
A Streamlit dashboard provides real-time visualization of the fleet, model convergence, and anomaly clusters to maintain operator interpretability.

- **Key Principles:**

1. **Multi-Exit Models:** Allows resource-constrained devices to train and contribute updates for shallower parts of the model, while powerful devices train deeper layers.
2. **Layer-wise Aggregation:** The fog robustly merges partial updates by averaging weights for each layer independently, using only the contributions available for that layer.
3. **Privacy-Preserving:** Only model weights are shared, never the raw sensor data.

ME-FLEET: RESULTS & ADVANTAGES



- **Improved Accuracy:** Achieved ~90% detection accuracy, a +50% improvement over the fuzzy model's ~60% .
- **Reduced Communication:** Sharing weights instead of raw data led to a ~40% reduction in communication overhead .
- **Adaptive Learning:** The CI model can adapt to new fault modes, unlike the static fuzzy system .
- **Collaborative Intelligence:** Creates a shared, fleet-wide intelligence model without centralizing sensitive data.

ME-FLEET successfully balances adaptability and performance while retaining interpretability through its visualization dashboard.

INDIVIDUAL CONTRIBUTION

Dwarakesh Venkatesh:

- Data generation
- Unsupervised Learning - Pairwise distance calculation
- Data visualization and model evaluation

Shyam Sundar Raju:

- Fog layer
- MQTT protocol

Sarigasini M:

- Unsupervised Learning - Agglomerative clustering
- Edge layer

Pooja Sree:

- Cloud layer
- Heartbeat mechanism